How Count Impacts Baseball

In which counts do players perform their best? Are some players more clutch than others?

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Methodology

1. Import Statcast data (2025 season) using pybaseball for all players
2. Construct plate appearances and label each by count (balls–strikes)
3. Compute per-player metrics (OBP, SLG, OPS) by count
4. Generate percentiles for each count for each player based on the range of OPS for each count
5. Aggregate league-wide totals by count (OBP, SLG, OPS) to create baseline performance
6. Generate graphics from data to be used for analysis

Terms to know:

OBP (On-base Percentage): Measures the rate at which a player reaches base for all other reasons than errors, fielder's choice, and dropped third strike

SLG (Slugging Percentage): Measures a batter's power by calculating the total bases per at-bat, not just hits

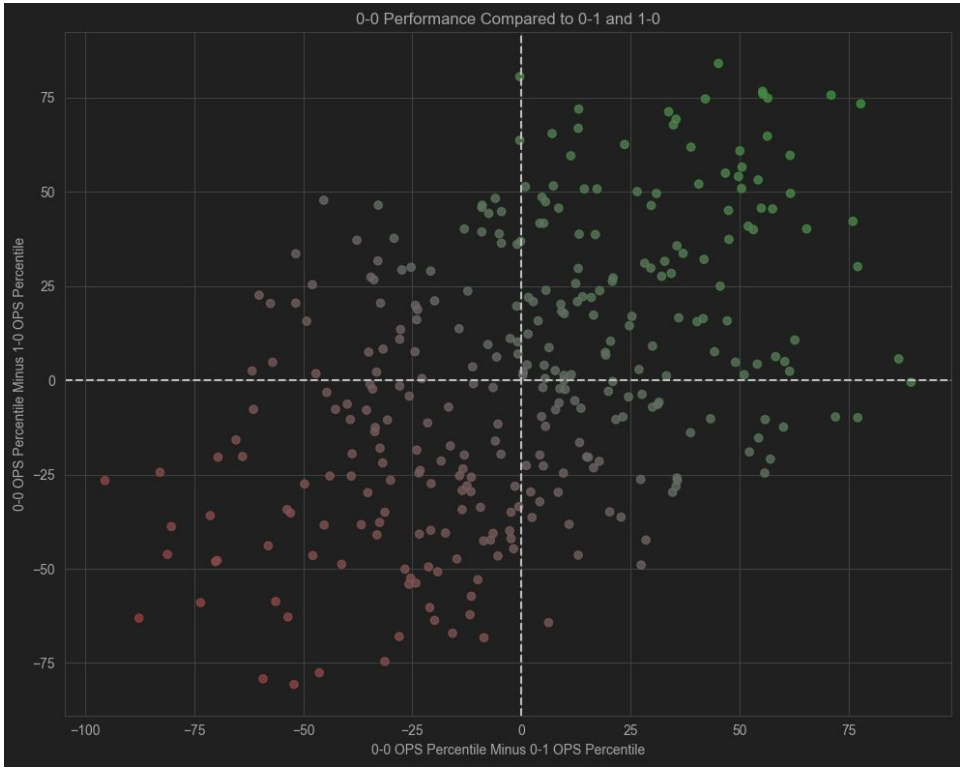
OPS (On-Base plus Slugging): Combination of the two metrics above, typically used as a measure of a hitter's total offensive production

How does swinging on 0-0 impact player performance?

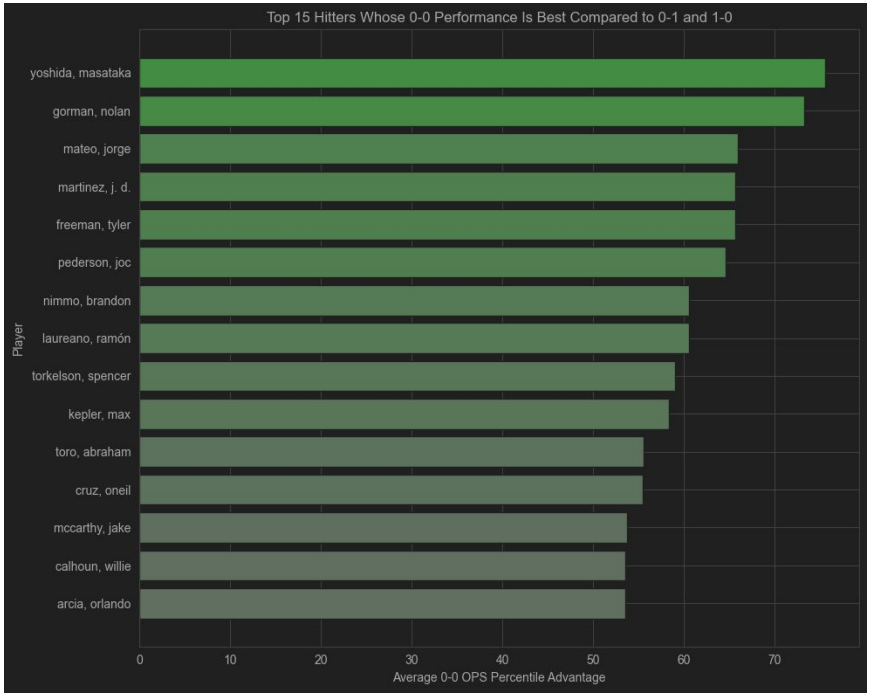
Actionability:

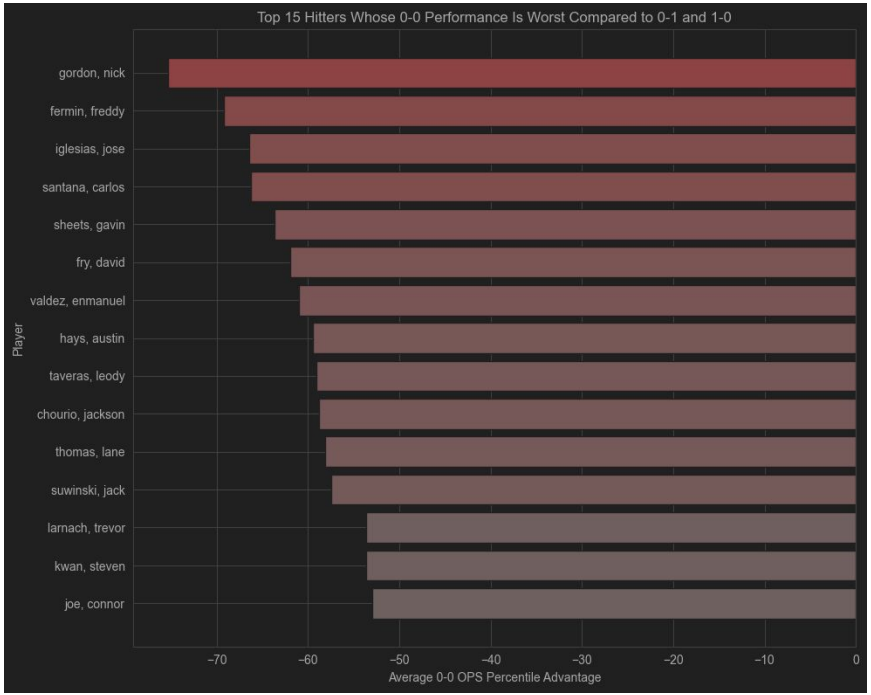
- Hitters who perform worse in 0-0 counts (bottom left) should be more selective on the first pitch.

- Hitters who perform better in 0-0 counts (top right) should be more aggressive on the first pitch.

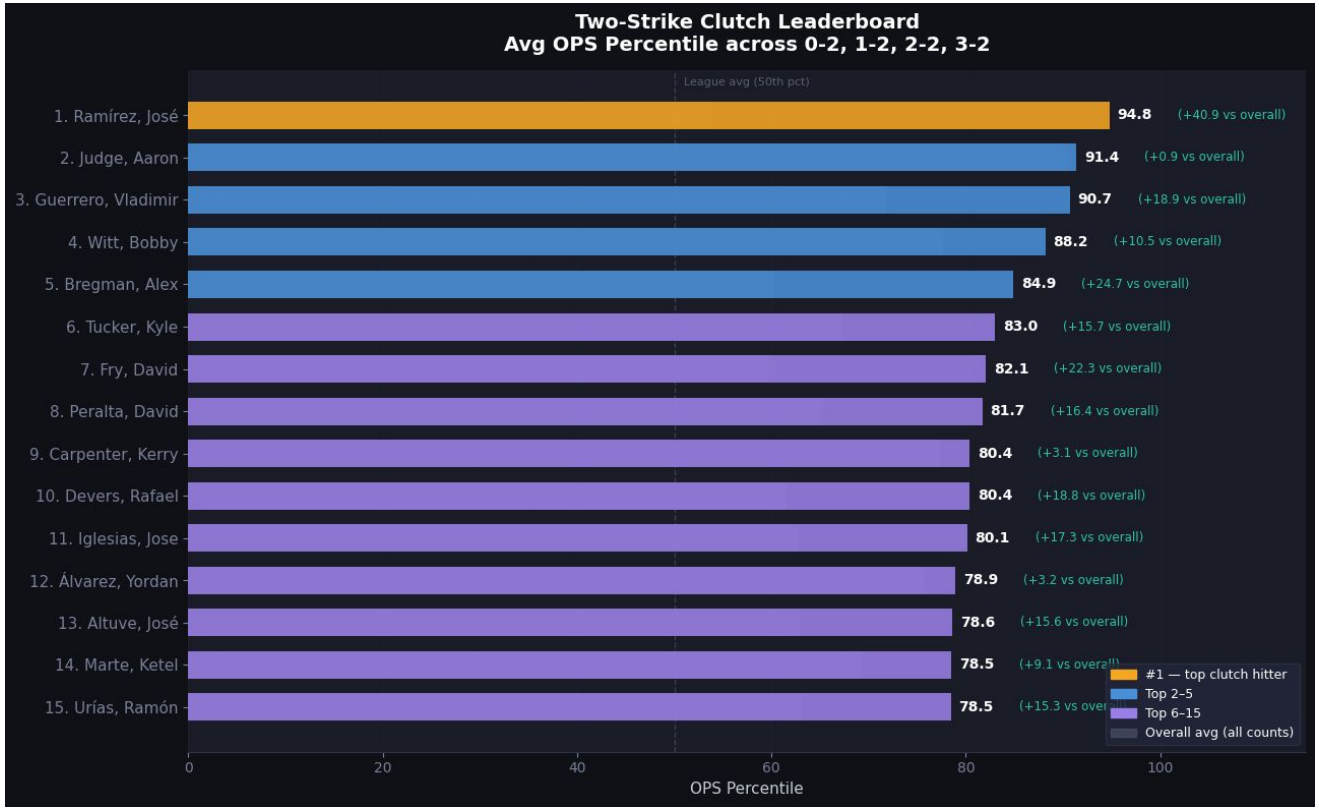


Which hitters benefit or struggle the most in 0-0 counts?



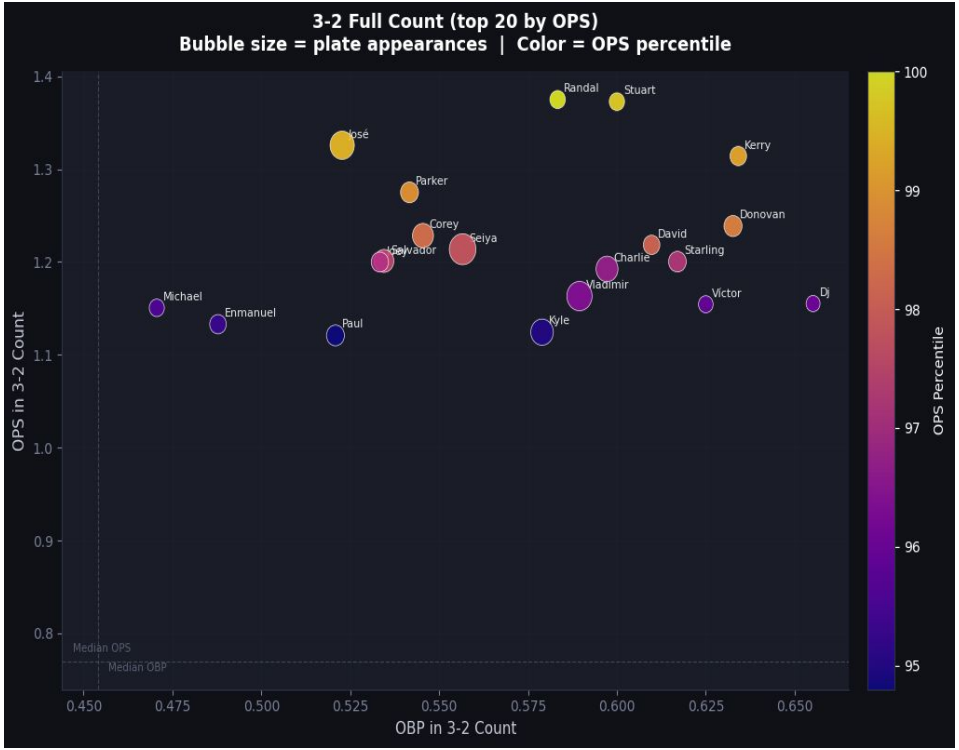


Who are the most clutch 2 strike hitters?

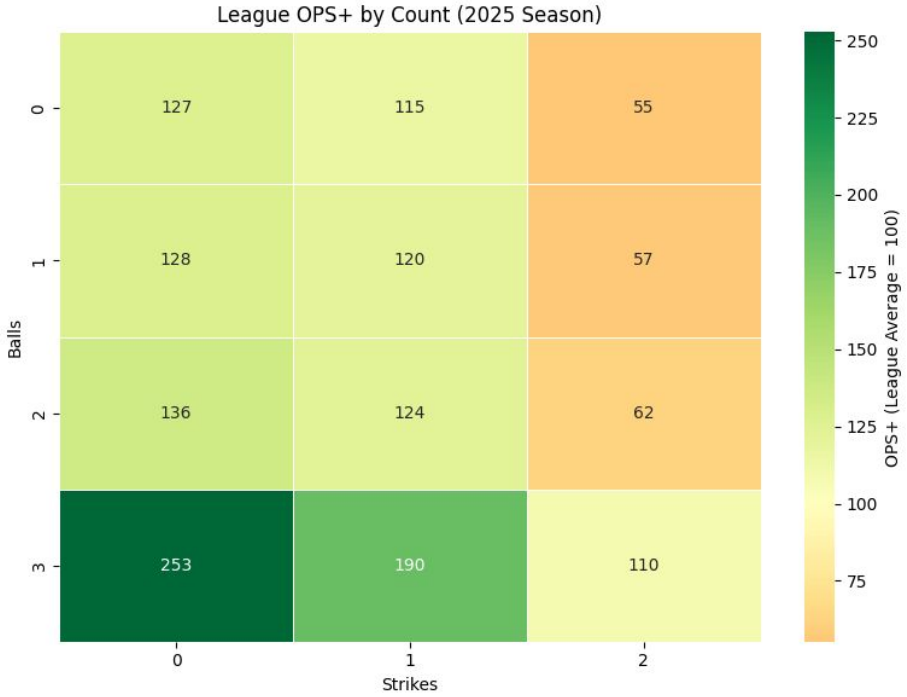


- Strong correlation between ops and clutchness at 2 strikes
- Data shows that the stronger hitters tend to also be the most clutch players
- **Actionable Insights (Batting Perspective)** -
    - Placing stronger hitters in spots 3-5 in the lineup seems to be the right move
    - Not only will they potentially get a big hit but if in a higher pressure situation they will be more likely to pull through for the team

- Strong correlation between ops, obp, and clutchness in the full count
- **Actionable Insight (Pitching Perspective)**
  - Expand the zone if you get to a full count on these batters
  - Clutch hitters are more likely to turn a 3-2 count into a hit versus an out
  - If it’s a close game and/or there are runners on base pitchers should consider throwing a pitch slightly outside the zone to give up a walk versus a big hit



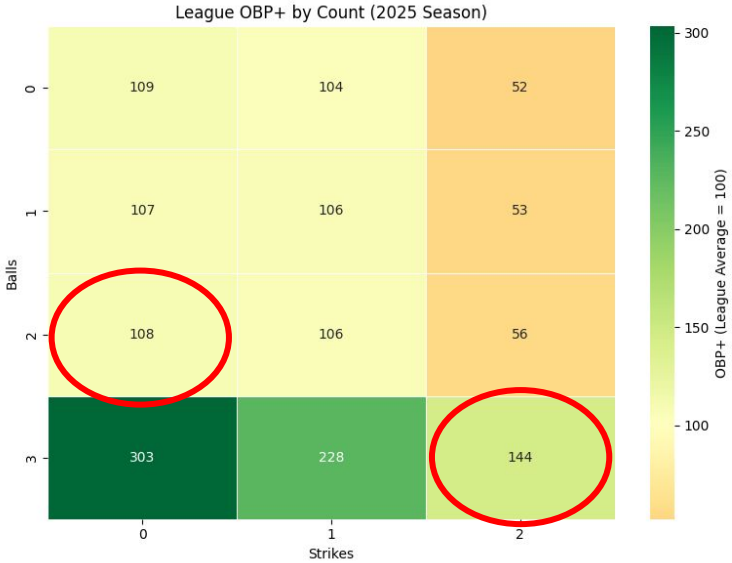
What does OPS by count really tell us?

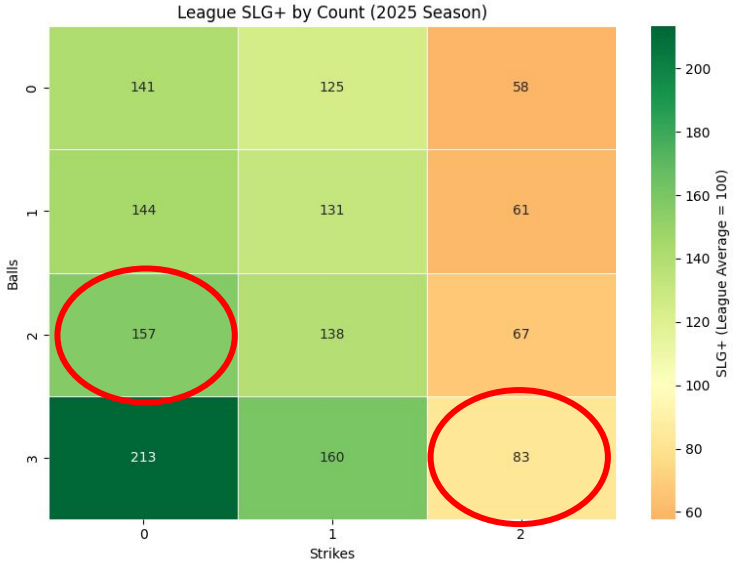


Insights:

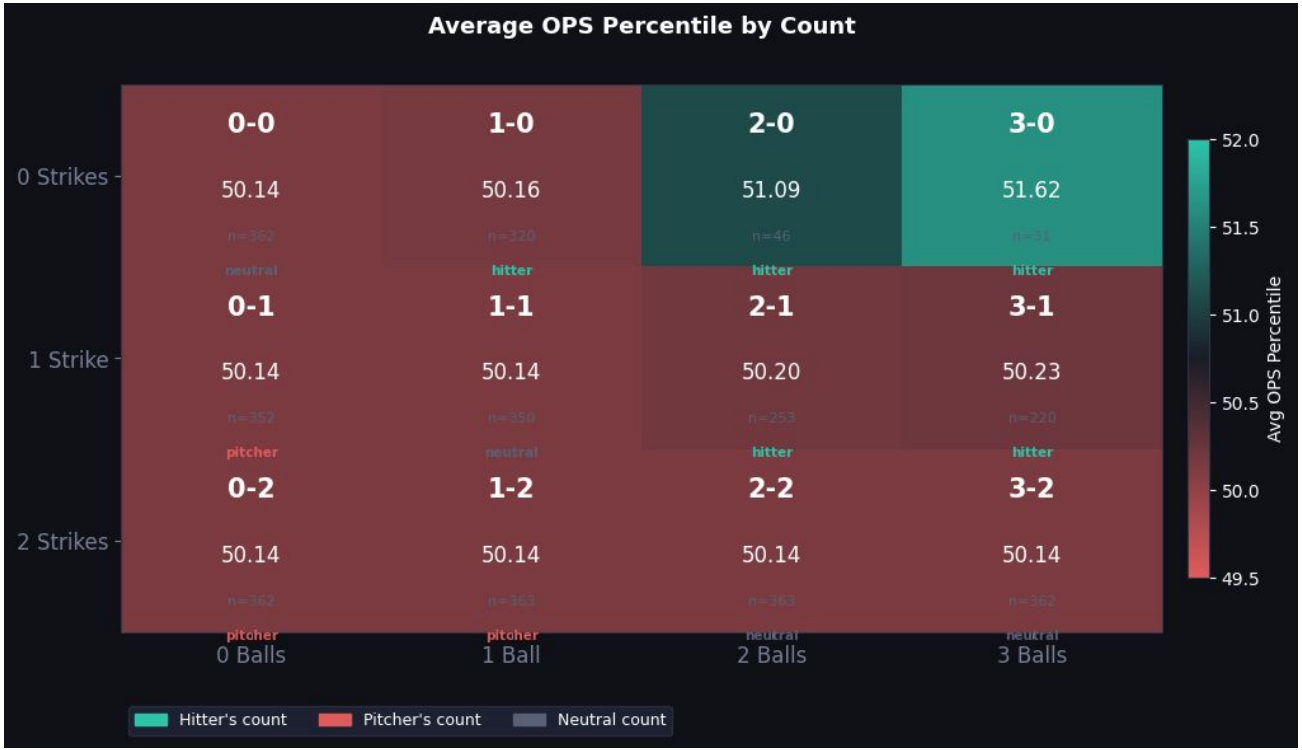
- Performance improves with more balls
- Two strike counts suppress offensive production
- Three ball counts favor well above-average outcomes

Breaking Down OPS: OBP vs SLG per count

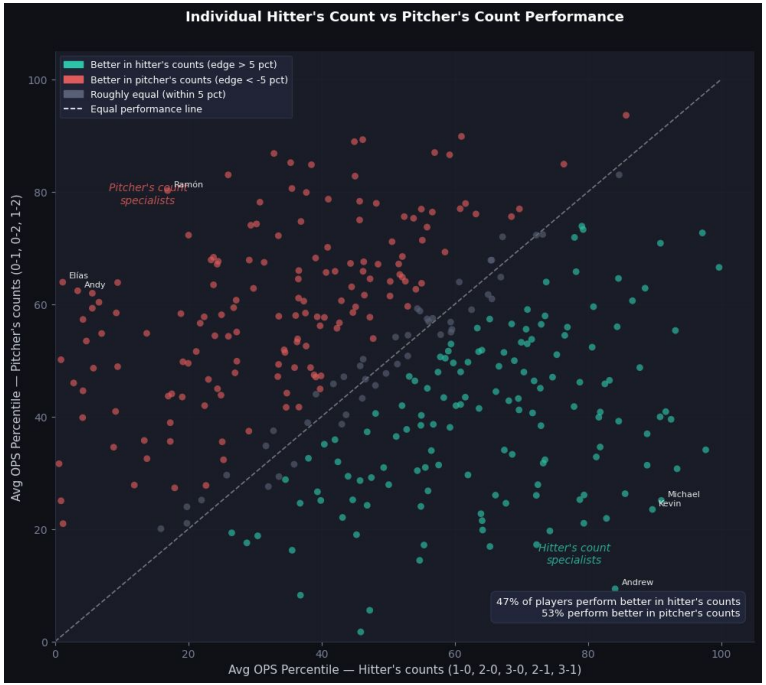




Do hitters actually perform better on hitter friendly counts?



- Varies batter to batter but overall batters have only a slight edge in batter friendly counts



- Players in the MLB who are better behind the count versus ahead in the count are roughly split fifty fifty
- 47% of players perform better on hitter friendly counts while 53% perform better on pitcher friendly counts
- Surprisingly batters do slightly better when the pitcher is ahead in the count

League wide there is no cohesive rule on how to handle different counts from a batting or pitching perspective but understanding how individual players perform at different counts can help pitchers understand how to attack them and help batting coaches understand how to guide their players on what to do in different counts